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Cohort study

THE PROGNOSIS OF ROOT CANAL THERAPY: A 20-YEAR FOLLOW-UP AMBISPECTIVE COHORT STUDY ON 411 PATIENTS WITH 1169 ENDODONTICALLY TREATED TEETH



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INTRODUCTION

Endodontic treatment aims to preserve the natural dentition in the oral cavity by preventing and treating apical periodontitis once endodontium contamination is established. Dental implants represent reliable substitutes when teeth are missing or teeth are severely compromised. However, the multiple therapeutic procedures (endodontic, periodontal, restorative and prosthetic treatments) involved present high biological and economic cost^{1,2}.

This raises important questions concerning the best treatment option for severely compromised teeth, and the long-term performance thereof^{1,3,4}.

Considering the endodontic prognosis alone, root canal therapy (RCT) is known to be a successful, highly reliable and non-invasive treatment⁵.

However, studies reporting the long-term outcomes (more than 10 years of follow-up) of endodontic therapy in severely compromised teeth and the factors influencing its effectiveness remain very rare, and the scientific evidence is still limited^{5,6}.

In particular, the impact of endodontic therapy on the overall long-term prognosis of teeth with restorative and prosthetic risk factors in patients susceptible to periodontal disease still remains undefined.

Therefore, the aim of this ambispective cohort study was to evaluate the outcomes of endodontic treatment over 20 years of follow-up in patients with advanced periodontal disease, and in teeth with multiple specific site/tooth risk factors, and to investigate the prognostic factors which could influence the long-term success of primary (treated for the first time) and secondary (re-treated) endodontic therapy.

MATERIALS AND METHODS

The present investigation is the continuation of a previously published study with 10-year follow-up data⁷. The difference between the two studies in the number of teeth included (1169 instead of 1175) is due to a calculation error detected in the interim. Specifically, it was noticed that in the first publication six teeth were counted twice in two patients (1 tooth extracted, 4 teeth healed, 1 tooth improved).

The study sample comprised any patient who had at least one tooth endodontically treated or re-treated by a single operator (F.F.) in a private practice between 1986 to 1998. The study was ambispective (retrospective until 2007 and prospective thereafter) prospectively.

The investigation adhered to STROBE Statement guidelines (www.strobe-statement.org) for observational studies. Informed consent was obtained from all participants included in the study after informing them about the procedures involved. In patients under 18 years of age, consent was obtained from a parent or legal guardian.

Endodontic therapy

Before endodontic therapy, periapical radiographs were taken of each tooth using the long-cone technique and a Rinn XCP film holder (Dentsply Sirona, Charlotte, NC, USA). For multi-rooted teeth, two preoperative periapical radiographs were taken from different angles to better evaluate root canal anatomy. Teeth with uncertain pulp vitality were subjected to sensitivity tests (cold, warm gutta-percha, and in a few cases cavity test).

Immediately before endodontic treatment, teeth were cleaned using an abrasive paste, isolated with a rubber dam, and disinfected with a cotton pellet soaked in 3% sodium hypochlorite. The pulp chamber was accessed using a water-cooled diamond bur (Intensiv 6916, Grancia, Switzerland). In root canal treatments performed before 1991, the working length was determined on periapical radiographs taken using the parallel technique after inserting a file into the canal, 0.5 mm coronally from the radiographic apex. Thereafter, an electronic apex loca-

tor was used (Neosono-D, Amadent Medical and Dental Co., Cherry Hill, NJ, USA; and after 1996, Root ZX, Morita Corp., Kyoto, Japan) at the "0" reading position.

Two preparation and obturation techniques were used: Schilder's technique between 1986 and 1993, and the crown-down technique from 1994 onwards⁸⁻¹⁰. For Schilder's technique^{8,9}, all canals were prepared with stainless steel files and root-filled with an adequate gutta-percha cone, adapted to the apical diameter, inserted with a small quantity of sealant (PCS, Kerr, Orange, CA, USA), and compacted vertically using heat carriers (Schilder spreaders size OP and OOP, Hu Friedy, Chicago, IL, USA). From 1991 onwards, Touch'n Heat was used for the purpose (Analytic Technologies, Redmond, WA, USA). Back-packing was performed manually in accordance with Schilder's technique prior to 1991⁸, and driven by Obtura syringe thereafter (Obtura Spartan, Foothill Ranch, CA, USA).

For the crown-down technique¹⁰, the coronal third of the canal was enlarged with Gates-Glidden drills (Maillefer, Balagues, Switzerland), while the remaining portion was filed using pre-curved stainless-steel Hedstrom files (Maillefer) with decreasing diameters from the middle to the apical third, without apical pressure. Standardized gutta-percha points were used to obturate the canals, choosing the cone with the diameter fitting the apical preparation. A cold lateral-compaction technique was used. The master cone was inserted into the canal with a small quantity of sealant (PCS); then accessory gutta-percha cones were laterally compacted with spreaders. Coronal excess of gutta-percha was removed, and vertically compacted using a heat carrier.

For both techniques, sodium hypochlorite 3% was used to rinse the canals after each instrument size. Ultrasound (EMS Piezon Master 400, EMS 1260, Nyon, Switzerland) was used to clean the canals for the crown-down technique only. In calcified canals, RT1 ultrasonic tips were used in association with ethylenediaminetetraacetic acid (EDTA) (RC PREP, Premier, Philadelphia, PA, USA) to remove calcifications. Canals were dried with sterile paper cones.

Upon completion of canal filling, the access cavity was temporized with Cavit W (3M ESPE, Seefeld, Germany), and definitively filled within 15 days to 1 month.

In all the teeth planned to be used as abutments for periodontal-prosthetic rehabilitations, the access cavity design and root canal preparation were very conservative to allow maximum preservation of tooth structure¹¹.

Between 1986 and 1993, calcium hydroxide was used, for 1 week in most cases, as intermediate medication in canals with persistent exudation. From 1994 onwards, treatment of symptomatic teeth, teeth with periradicular radiolucencies, and canals with exudation was completed over the course of two appointments. Calcium hydroxide paste was used, mainly for 1 week, as medication between endodontic sessions. In re-treated teeth with radiolucencies, the dressing was performed with iodoform paste (Ogna Lab, Muggiò, Italy) for 1 week.

Follow-up protocol

All patients were recalled for a scaling/root planing session every 3, 4 or 6 months, depending on the periodontal and dental risk factors. The treating clinician (F.F.) checked the clinical conditions of the treated teeth annually. The first radiographic control was performed at 1 year for all endodontically treated teeth (at 6 months only for teeth that had been symptomatic following treatment and/or displayed preoperative periradicular radiolucency). Following the first year, and up to the final 20-year evaluation, radiographic check-ups were scheduled every two years or more, in line with ADA recommendation for patients with multiple risk factors¹².

All patients were invited to attend the 20-year post-endodontic treatment evaluation. The reason for and date of drop-outs were recorded, as well as the date of the last clinical/radiographic assessment for the patients who did not attend the final 20-year follow-up appointment.

Outcomes

The outcomes recorded were the following.

- Tooth survival, as assessed by the treating clinician (F.F.) during the follow-up check. Reasons for any failure were recorded. Extracted teeth or roots were examined under 4.5x magnification, and in the event of persistent diagnostic doubts, coloured with methylene blue to disclose vertical root fractures.
- Treatment success, as assessed clinically by the treating clinician (F.F.) and radiologically by one blinded independent and experienced assessor (P.B.) in the following way: complete success (asymptomatic tooth with complete absence of visible periradicular radiolucency in periapical radiographs); partial success (reduced radiolucency but tooth asymptomatic and in perfect function); partial failure (not reduced/worsened radiolucency or radiolucency not present before the treatment but tooth still asymptomatic and perfectly functional), and failure (tooth that needed to be extracted for endodontic or other reasons¹³. Multi-rooted teeth were scored according to the root having the worst outcome.
Periapical radiographs taken up to 2007 were examined on a radiograph dental viewer using a magnifying lens with 4x magnification in a partially darkened room. Thereafter, digital periapical radiographs were taken and evaluated on a computer screen. Prior to the radiographic evaluation, the blind independent assessor was calibrated.
- Complications, as assessed by the treating clinician (F.F.). Any endodontic or suspected endodontic complication, such as periapical abscess or clinical situations resembling acute endodontic infection with sulcular drainage and rapid increase in probing occurring over the 20-year follow-up period was recorded and is reported. When the complication precluded maintaining the tooth, the consequent extraction was recorded as a failure, and no longer considered a complication. Successfully treated complications on teeth subsequently extracted for other reasons than the complication itself have also been reported.

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CONCLUSIONS

The 20-year prognosis of endodontically treated/re-treated teeth as part of multidisciplinary rehabilitation of patients affected by advanced periodontitis is good. Twenty years after endodontic therapy, clinical survival rate was 86%, with 12.9% of teeth being extracted, only 0.9% for endodontic reasons, and 2.6% of teeth affected by successfully treated complications.

Aging, endodontic treatment in two sessions, and endodontic treatment for non-endodontic reasons are important predictors of failure, while preoperative radiolucency and endodontic re-treatment seem not to be. Root canal therapy can be considered a successful treatment long-term, even for teeth with multiple risk factors or those used as abutments for perio-prosthetic rehabilitations.

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